# Sourabh Joshi

VNIT Nagpur, India | joshisourabh47@gmail.com | +91 96600 88773
sourabhjoshi.info | linkedin.com/in/sourabhk-nitb | github.com/sourabhjoshi1

## Professional Summary

M.Tech Electrical Engineering student at VNIT Nagpur with experience in power electronics, electrical drives, battery management systems, control systems, and industrial electrical maintenance. Skilled in MATLAB/Simulink, Python, machine learning, DC-DC converters, Li-ion battery SoC/SoH estimation, SiC/GaN wide-bandgap devices, and power conversion systems.

## Education

**Visvesvaraya National Institute of Technology (NIT Nagpur)** July 2024 – Expected June 2026
*M.Tech in Electrical Engineering* *CGPA: 7.22/10*

- **Relevant Coursework:** Power Electronics, Electrical Drives, Control Systems, Battery Management Systems

**Maulana Azad National Institute of Technology (NIT Bhopal)** July 2017 – May 2021
*B.Tech in Electrical Engineering* *CGPA: 6.91/10*

## Experience

**MSP Steel and Power Limited** Jan 2022 – Jan 2023
*Graduate Engineering Trainee, Pellet Division – Electrical Department* *Raigarh, Chhattisgarh*

- Supported operation, preventive maintenance, and troubleshooting of electrical systems in a pellet plant environment.
- Worked with motor control centers (MCCs), transformers, automation systems, motors, starters, and plant electrical distribution equipment.
- Assisted in identifying electrical faults, coordinating corrective actions, and maintaining reliable operation of production equipment.

## Projects

**AI-Based State-of-Charge and State-of-Health Estimation for Li-Ion Batteries** July 2025 – Present

- Developing a deep learning framework for Li-ion battery State of Charge (SoC) and State of Health (SoH) estimation using LSTM and hybrid machine learning models.
- Working with battery data preprocessing, feature extraction, model training, validation, and performance evaluation for battery management system applications.
- Applying Python-based machine learning methods to improve estimation accuracy under varying battery operating conditions.

**Performance Comparison of Si, SiC, and GaN-Based High-Frequency DC-DC Converters** Dec 2025 – Present

- Analyzing high-frequency DC-DC converter performance using silicon, silicon carbide, and gallium nitride power semiconductor devices.
- Evaluating efficiency, switching losses, thermal performance, power density, and system cost for wide-bandgap power electronics applications.
- Comparing device-level and converter-level tradeoffs relevant to modern power conversion systems.

**BLDC Motor Speed Control Using MATLAB/Simulink** Jan 2020 – May 2020

- Developed a closed-loop speed control model for a brushless DC motor using a PID controller in MATLAB/Simulink.
- Improved transient response by reducing steady-state error and settling time through controller tuning and simulation analysis.

## Publication

**A Comparative Analysis of Various Deep Learning Models for State-of-Charge Estimation in Lithium-Ion Batteries**

- Presented at the IEEE 4th International Conference on Smart Technologies for Power, Energy, and Control (STPEC 2025), hosted by NIT Goa, Dec 10–13, 2025.

## Certifications

**Supervised Machine Learning: Regression and Classification**, DeepLearning.AI May 2025
Coursera Certificate

**Introduction to Power Electronics**, University of Colorado Boulder Mar 2025
Coursera Certificate

**Arena FIDE Master**, FIDE Dec 2021

## Technical Skills

**Programming Languages:** Python, MATLAB, C
**Simulation and Analysis:** MATLAB/Simulink, machine learning, deep learning, LSTM models, data preprocessing, model validation
**Electrical Engineering:** Power electronics, DC-DC converters, inverters, electrical drives, control systems, motor control, BLDC motors
**Battery and Energy Systems:** Li-ion batteries, battery management systems, battery energy storage systems, SoC estimation, SoH estimation
**Power Devices:** Si, SiC, GaN, wide-bandgap semiconductor devices, switching losses, thermal performance, power density
**Industrial Systems:** MCCs, transformers, motors, automation systems, electrical troubleshooting, preventive maintenance

## Achievements and Memberships

- GATE 2026: All India Rank 4032
- GATE 2024: All India Rank 5151
- JEE Main 2017: All India Rank 14780
- IEEE Graduate Student Member and IEEE Power Electronics Society (PELS) Member
- Member of FIDE and CASP16